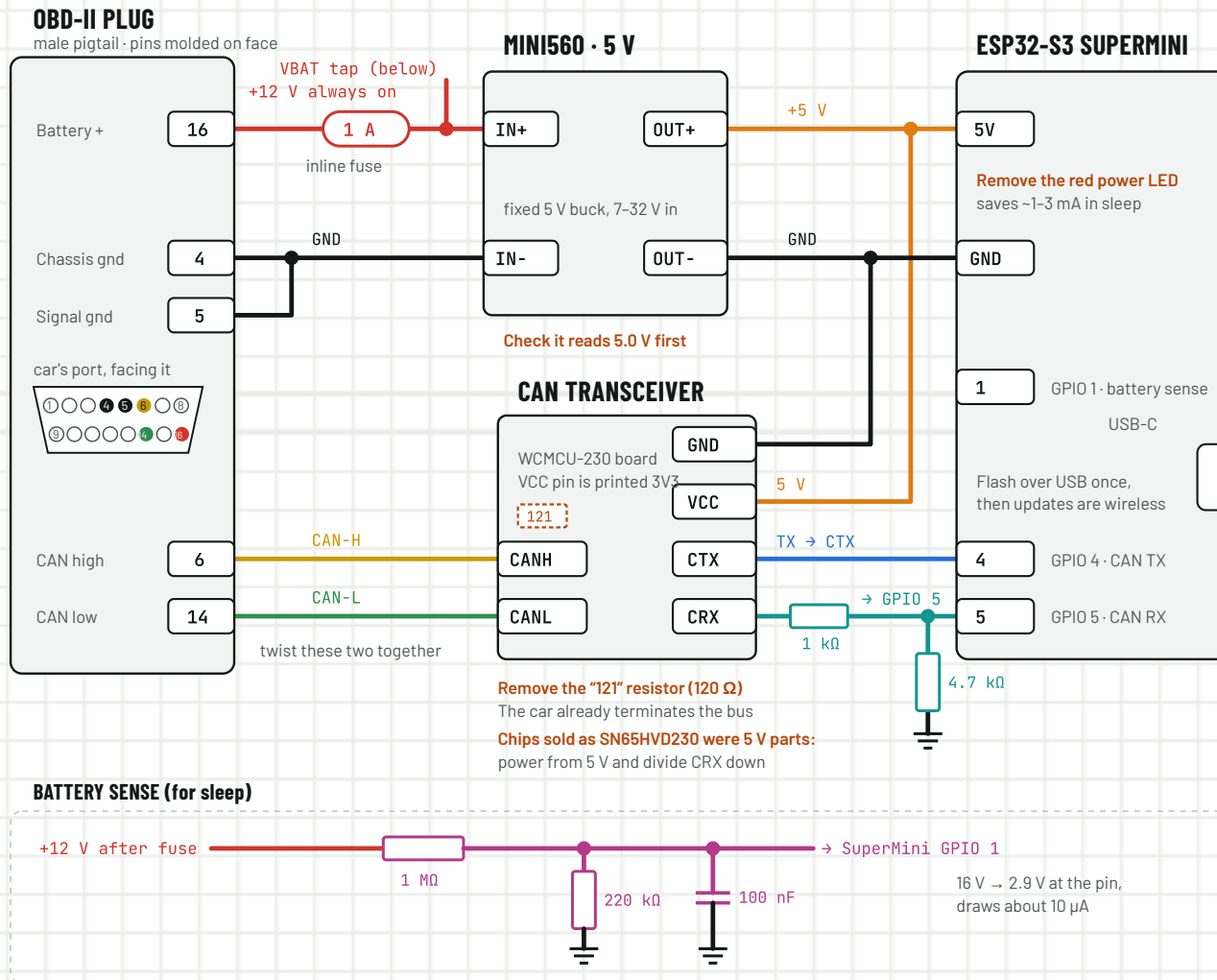


CAN logger wiring

Always-on power from OBD pin 16 through a 1 A fuse and a fixed 5 V Mini560 into an ESP32-S3 SuperMini. A CAN transceiver, powered from 5 V, connects the S3 to the diagnostic CAN bus on pins 6 and 14; its receive line is divided down for the S3. A battery-voltage divider lets the board sleep while the car is off. Pin labels are drawn as they are printed on each board, not where they sit physically.



— 12 V (fused) — 5 V — Battery sense — Ground — CAN TX — CAN RX — CAN-H — CAN-L

Wire list

FROM	TO	NOTES
 OBD 16	Fuse → Mini560 IN+	1 A fuse close to the plug
 OBD 4 + 5	Mini560 IN-	Join both pins
 Mini560 OUT+	SuperMini 5V and transceiver VCC	Transceiver's pin is printed 3V3
 Mini560 OUT-	SuperMini GND	
 Transceiver GND	SuperMini GND	Same ground point
 SuperMini GPIO 4	Transceiver CTX	Direct
 Transceiver CRX	1 k Ω → GPIO 5	4.7 k Ω from GPIO 5 to GND; check ≤ 3.4 V
 Transceiver CANH	OBD 6	Twisted pair, keep it short
 Transceiver CANL	OBD 14	
 +12 V after fuse	1 M Ω → GPIO 1	220 k Ω and 100 nF from GPIO 1 to GND

22–24 AWG is plenty for every wire here. The whole board draws well under half an amp.

Build order

- 1 Check the Mini560 on the bench**
Feed it 12 V and measure OUT+ to OUT-. It should read about 5.0 V. A mislabeled 9 V or 12 V module would destroy the S3.
- 2 Prep the boards**
Remove the “121” resistor from the transceiver and the red power LED from the SuperMini.
- 3 Check the CRX divider**
Power the transceiver from 5 V with the divider built but not yet on GPIO 5. The junction should read 2.9–3.4 V.
- 4 Solder the harness**
Follow the wire list. Heat-shrink every joint and twist CAN-H and CAN-L together.
- 5 Flash over USB-C**
Do this with the harness unplugged from the car, so the computer and the Mini560 never power the board at the same time.
- 6 Plug into the car**
With the ignition on, the status page should show “tx ok” climbing and frames received.
- 7 Let sleep arm itself**
The board only starts sleeping once its battery reading matches the engine computer's. The status page says when.
- 8 Measure sleep current**
Put a multimeter in series with the 12 V wire after the car locks and goes quiet. Under 3 mA is the target.

BEFORE PLUGGING IN

- CANH and CANL swapped won't damage anything, but the board will see no traffic.
- CRX swings to about 3.7 V on these boards; never connect it to GPIO 5 without the divider.
- A genuine 3.3 V SN65HVD230 would go on the 3V3 pin with CRX direct instead.
- Don't power the board from USB and the car at the same time.